

Financial Markets as 0/0 Removable Singularities

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2026

The Removable Singularity Framework

Abstract

We show that financial markets exhibit 0/0 removable singularities at critical points: the Black-Scholes boundary ($V = 0/0$ at $S=0$ or $T=0$), market crashes (Gaussian to power-law phase transition), Sornette's log-periodic model (LPPLS for crash prediction), Mandelbrot's fractal finance (universal Hurst exponent), herding as Kuramoto synchronization (same model as consciousness), and GARCH unit root (IGARCH at $\alpha+\beta=1$). The key insight: human markets obey the SAME universal 0/0 laws as physics. The financial 0/0 connects to Ising (phase transitions), Kuramoto (consciousness), Kolmogorov (fractals), and Eigen (error threshold).

1. Black-Scholes Boundary (0/0)

$$V = S \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$$

At $S=0$: $V = 0$ (removable). At $T=0$: $V = \max(S-K, 0)$ (removable). At $\sigma=0$: $V = \max(S - Ke^{-rT}, 0)$ (removable). The option price approaches a finite limit from all directions -- the hallmark of a removable singularity.

2. Market Crash as Phase Transition

Normal market: returns are Gaussian (efficient market hypothesis). Critical market: returns become power-law distributed (herding, bubbles). At the crash: 0/0 removable singularity.

- Pre-crash: log-periodic oscillations (Sornette 1999)
- Crash: power-law divergence in price
- Post-crash: relaxation to new equilibrium

3. Sornette Log-Periodic Model

$$\ln(P(t)) = A + B \cdot (t_c - t)^m + C \cdot (t_c - t)^m \cdot \cos(\omega \ln(t_c - t) - \phi)$$

At $t = t_c$: price diverges (0/0 removable singularity). Log-periodic oscillations BEFORE the crash are the signature of the approaching singularity. This model successfully predicted the 2008 financial crisis (Sornette & Johansen, 2001).

4. Mandelbrot Fractal Finance

Mandelbrot showed that financial returns are FRACTAL, not Gaussian. The Hurst exponent H is UNIVERSAL across all markets:

- $H = 0.5$: Brownian motion (efficient market)
- $H > 0.5$: persistent trends (momentum)
- $H < 0.5$: mean-reverting (anti-persistent)

The fractal dimension $D = 2 - H$. For turbulence: $H = 1/3$ (Kolmogorov). For efficient markets: $H = 0.5$. The critical point $H = 0.5$ is a 0/0: below, mean-reverting; above, trending.

5. Herding as Kuramoto Synchronization

Traders can be modeled as coupled oscillators (Kuramoto model): opinion = phase angle, social influence = coupling K ,

independent analysis = noise. Below critical K: independent trading (efficient). Above critical K: herding (synchronized, unstable). At K_c : 0/0.

This is the EXACT SAME model as consciousness (Chapter 34)! The market crash is a loss of "consciousness" -- the market loses its ability to process information independently.

6. GARCH Volatility and Unit Root

$$\sigma_t^2 = \omega + \alpha \cdot r_{t-1}^2 + \beta \cdot \sigma_{t-1}^2$$

At $\alpha + \beta = 1$: unit root (0/0 removable singularity). The IGARCH model has INFINITE volatility -- the market becomes unstable. This is the financial analogue of the Toomre $Q = 1$ threshold in galaxies.

7. Connections to All Prior Work

Finance	Physics	0/0 Type
Black-Scholes boundary	Ising phase transition	Boundary
Market crash	Eigen error threshold	Sudden
Herding	Kuramoto synchrony	Collective
Mandelbrot fractal	Kolmogorov $-5/3$	Scaling
GARCH unit root	Toomre $Q=1$	Stability
Var threshold	Bekenstein bound	Information

8. Conclusion

Financial markets are 0/0 removable singularities. The Black-Scholes boundary, market crashes, Sornette LPPLS, Mandelbrot fractals, herding as Kuramoto, and GARCH unit root all share the same 0/0 structure as phase transitions in physics. Human markets obey the SAME universal laws as the natural world.

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